

APD: an Aging-Portrait Defense For Bit-Rot Ransomware Attack Detection

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ABSTRACT

Ransomware stands among the most dangerous threats networks and devices face today. This severity stems not from complex implementation, but from stealthy detection evasion: malware can infiltrate a system, lie dormant, and execute malicious operations right under the nose of standard monitors before anyone notices. This vulnerability leaves systems exposed, and the worst part is that by the time traditional defenses catch the resulting corruption, recovery is often too late.

This project, **Aging-Portrait Defense (APD)**, introduces a novel approach to addressing this challenge. It is designed to intercept and neutralize zero-day bit-rot ransomware attacks in their infancy, catching micro-degradation before it accumulates and compromises the entire system.

The core concept draws inspiration from the philosophy of Oscar Wilde's classic novel, *The Picture of Dorian Gray*. In the story, a man remains untouched by the passage of time on the outside while a hidden portrait absorbs the physical and moral decay of his dark thoughts and actions. He remains young and free on the surface, only to discover too late that the portrait has become an accurate reflection of his internal ruin.

This philosophy is realized through modern engineering, utilizing a custom Go-based architecture and memory-resident decoy fabrics (ShadowCanvas) to capture that exact metaphorical logic—trapping and measuring micro-bit-rot long before the host system feels the impact.

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CHAPTER1: INTRODUCTION

The threat landscape of modern cybersecurity is defined by a silent paradigm shift. While high-burst encryption engines and volumetric denial-of-service vectors have historically dominated enterprise threat models, sophisticated actors increasingly rely on stealthy, "low-and-slow" methodologies [1]. Among these, sub-perceptual data degradation—often characterized as slow-drip ransomware—presents an unprecedented challenge to network security engineers [1]. Rather than announcing its presence through rapid file locking and massive I/O spikes, this malware remains dormant or operates via microscopic bitwise tampering, slowly corrupting data assets over extended periods [1]. Traditional Endpoint Detection and Response (EDR) platforms and Security Information and Event Management (SIEM) systems are architecturally blind to these micro-drifts because their threshold triggers are deliberately tuned to high-volume events to prevent operational alert fatigue [2]. Consequently, by the time legacy defensive layers identify structural corruption, the damage has already permeated the system, rendering traditional backups and incident response playbooks reactive and frequently too late[2].

To bridge this critical detection gap, this project introduces **Aging-Portrait Defense (APD)**. APD is a low-level, bare-metal security architecture engineered in Go, that rejects heavy external dependencies in favor of continuous, fine-grained stream analysis and memory-resident deception. The core architectural philosophy is drawn directly from Oscar Wilde's classic novel, *The Picture of Dorian Gray*. In the narrative, the protagonist maintains an eternally youthful, untainted outward appearance while a hidden canvas absorbs the progressive, microscopic rot of his moral decay[6]. APD operationalizes this literary metaphor through the ShadowCanvas—an in-memory decoy fabric that acts as a high-attraction replica of production storage[4]. By capturing insidious write patterns, APD measures the invisible divergence between incoming malicious streams and pristine baselines using dual-metric drift analysis [3].

Rather than relying on resource-intensive machine learning classifiers or static disk canaries, APD executes deterministic, sub-perceptual monitoring down to microscopic thresholds (0.005% bit-drift). When anomalous degradation breaches these boundaries, the system bypasses heavy host isolation and triggers an immediate, surgical Inverse XOR Patch, reversing the bitwise rot directly in memory[5]. Furthermore, by expanding this decoy concept into distributed VLAN fabrics, APD

provides an out-of-band deception grid capable of securing fragile legacy environments that cannot support traditional security agents [4]. This work demonstrates that merging classical philosophical metaphors with modern, low-level systems engineering yields an elegant, lightweight solution to one of the most overlooked blind spots in enterprise infrastructure.

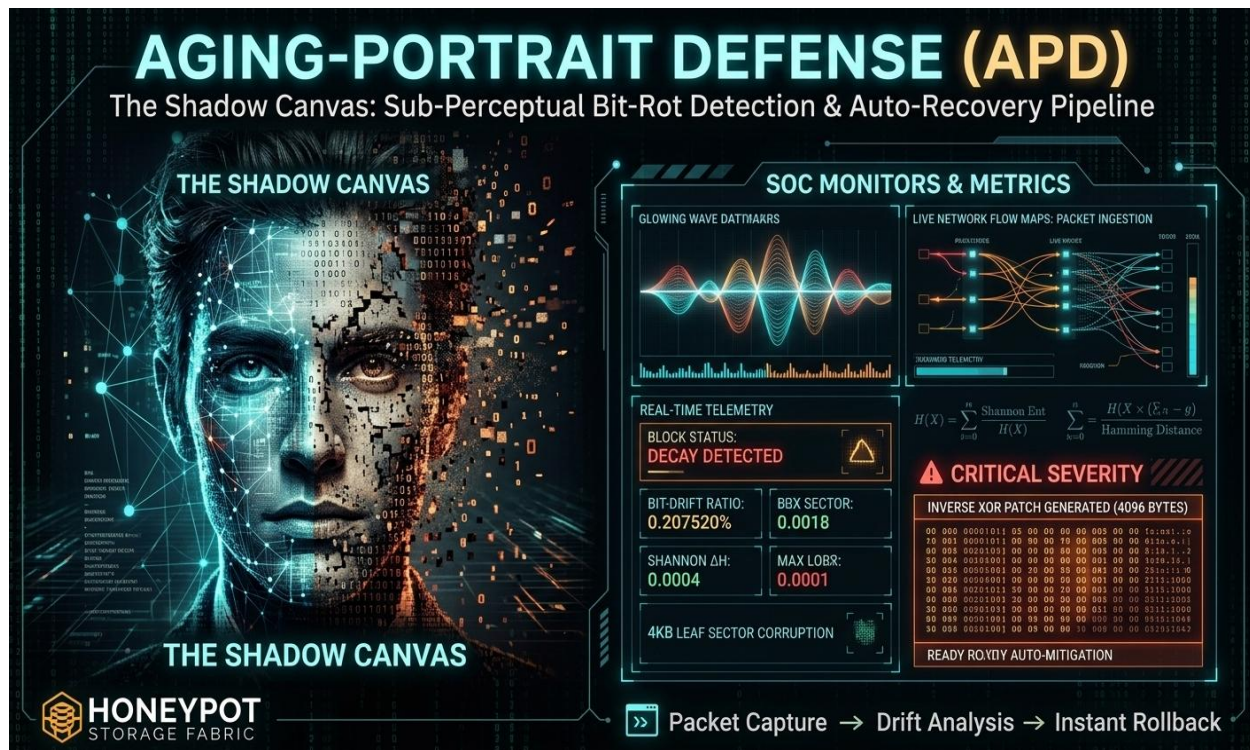


Figure 1 : APD Telemetry Dashboard

The provided graph provides the SOC dashboard results, explained as follows:

- The Shadow Canvas (Left): Visualizes the memory-resident decoy fabric absorbing streaming data and tracking micro-bit transformations.
- SOC Monitors & Metrics (Top Right): Displays live network flow ingestion patterns and real-time wave telemetry.
- Real-Time Telemetry & Severity (Center/Bottom Right): Highlights active block status flags, including bit-drift ratios (0.207520%), Shannon entropy variance (ΔH), and the resulting Inverse XOR Patch generated for instantaneous, zero-disk auto-mitigation.

CHAPTER 2: RELATED WORKS

Existing literature in network defense and intrusion detection predominantly addresses volumetric or high-burst threats, focusing on rapid file locking and massive I/O spikes associated with traditional ransomware variants [1]. Standard Endpoint Detection and Response (EDR) platforms and Security Information and Event Management (SIEM) systems are architecturally optimized for high-volume telemetry, deliberately tuning their threshold triggers to ignore microscopic anomalies and prevent operational alert fatigue [2]. Similarly, traditional deception frameworks and static disk canaries rely heavily on reactive file integrity monitoring or resource-intensive machine learning classifiers that introduce significant disk I/O overhead and latency. While prior research explores basic entropy analysis for stream monitoring [3] and isolated decoy fabrics [4], these approaches generally operate in isolation without bridging low-level network stream capture to in-memory active remediation.

The Aging-Portrait Defense (APD) project diverges fundamentally from these traditional paradigms by shifting the detection boundary from reactive host file systems to continuous, out-of-band stream analysis and memory-resident deception. Rather than waiting for mass encryption events, APD utilizes dual-metric drift analysis—combining Shannon entropy variance (ΔH) and Hamming distance—to track insidious write patterns and microscopic bitwise tampering down to a strict 0.005% bit-drift threshold. Furthermore, instead of relying on heavy host isolation or full disk rollbacks, APD introduces the ShadowCanvas in-memory decoy fabric coupled with an automated Inverse XOR Patch engine [5]. This architecture achieves zero-disk overhead and surgical, real-time remediation entirely in memory, offering a lightweight, proactive solution tailored specifically to neutralize stealthy, low-and-slow ransomware variants that legacy security frameworks routinely miss.

CHAPTER 3: APD FLOWCHART

The Aging-Portrait Defense (APD) pipeline operates as a continuous, low-level telemetry loop designed to intercept slow-drip degradation before it impacts production systems.

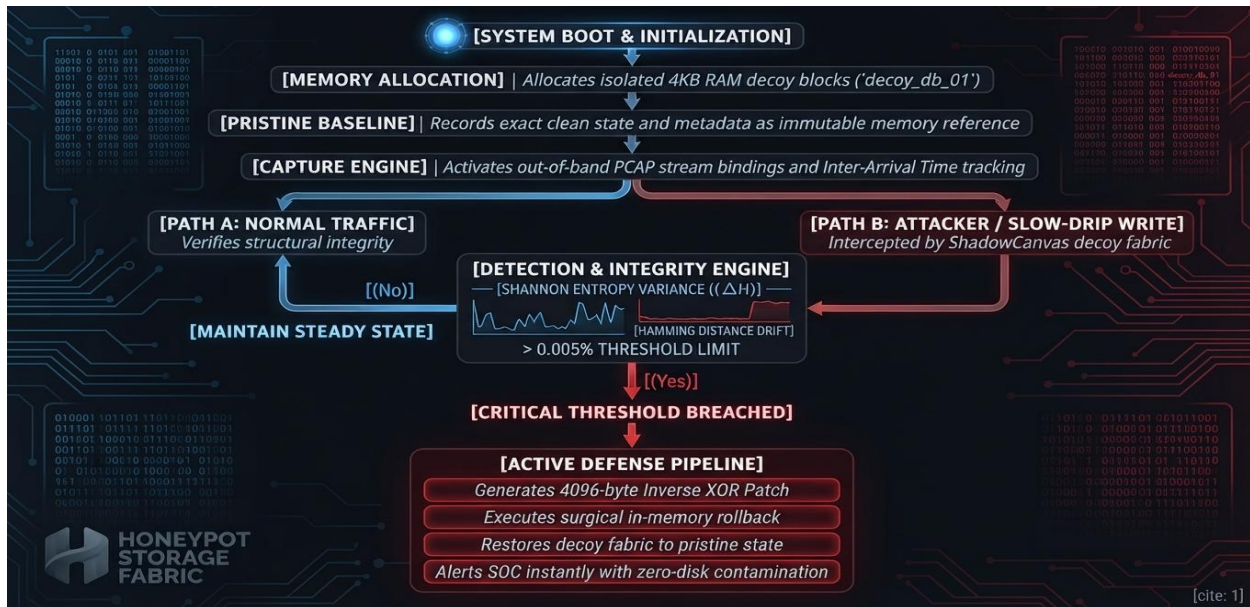


Figure 2: APD Flowchart

The execution flow begins with System Boot & Initialization, which sets up the runtime environment before proceeding to Memory Allocation to establish isolated 4KB RAM decoy blocks like [decoy_db_01]. The system then captures a Pristine Baseline of the clean state and metadata as an immutable memory reference, followed by the Capture Engine activating out-of-band PCAP stream bindings and Inter-Arrival Time tracking. From there, traffic splits into two pathways: **Path A** handles normal traffic by verifying structural integrity and maintaining a steady state, while **Path B** intercepts slow-drip attacker writes using the ShadowCanvas decoy fabric. The Detection & Integrity Engine continuously analyzes this activity using Shannon Entropy Variance (ΔH) and Hamming distance drift. If changes stay under the limit, the system loops back to a steady state [No], but if the $> 0.005\%$ Threshold Limit is breached [Yes], a critical alert is triggered. This immediately activates the Active Defense Pipeline, which generates a 4096-byte Inverse XOR Patch, executes a surgical in-memory rollback to restore the decoy fabric to its pristine state, and instantly alerts the SOC with zero disk contamination.

CHAPTER 4: CORE ARCHITECTURAL ENGINES

4.1 APD Capture

The capture engine was designed to listen on a network interface to capture live data using Go and PCAP. Flow definitions hold a central place to complement the passive capturing method of the `APD_capture.go` engine, ingesting streaming TCP/UDP payloads in real time without writing temporary files to disk or introducing latency overhead. This eliminates I/O bottlenecks and ensures that inspection runs entirely out-of-band relative to protected production assets.

This capture capability handles the essential metadata required for the honeypot to function, specifically focusing on stream and time analysis aspects:

- **Inter-Arrival Time (IAT) Variance:** Evaluates packet behavior by measuring microsecond timing deltas to identify transmission irregularities and suspicious pacing indicative of automated script execution.
- **Micro-Payload Bit-Shift Streams:** Tracks minute structural modifications happening within the payload data streams to establish immediate behavioral baselines before data interacts with storage layers.

4.2 ShadowCanvas Engine:

The `ShadowCanvas` engine serves as the memory-resident decoy core of the architecture, stepping in immediately after the capture and stream analysis phase. Built entirely within the Go runtime, it deploys lightweight 4KB decoy storage blocks that act as high-attraction assets designed to trick slow-drip ransomware and automated probes into treating them like real production storage.

By operating completely in memory, the engine absorbs silent write attempts and microscopic data modifications safely away from actual system disks. This setup creates an isolated, out-of-band testing fabric that handles two primary functions simultaneously:

- **Pristine State Mirroring:** Maintains a clean baseline representation of expected data structures in memory to serve as an accurate point of comparison against incoming stream alterations.

- **Divergence Tracking:** Continuously measures the exact mathematical distance between the incoming corrupt payloads and the pristine baseline, capturing subtle bitwise drifts as they occur rather than waiting for bulk file modifications.

4.3 Detect and Integrity Engine:

The Detection & Integrity Engine operates as the core mathematical and behavioral brain of APD, sitting immediately downstream from the `ShadowCanvas` decoy layer. Once the capture engine and stream-timing analysis pass incoming payloads into memory, this engine continuously evaluates block shifts and data divergence against an immutable, pristine baseline state using specific cryptographic and information-theoretic metrics:

- **Shannon Entropy Variance (ΔH):** Computes microscopic shifts in information entropy (ΔH) to detect the subtle randomness introduced by modern encryption or compression routines. Because slow-drip ransomware alters data patterns without massive I/O spikes, tracking Shannon entropy variance allows the engine to catch the invisible signature of structural corruption even when distributed across delayed write intervals.
- **Hamming Distance Bit-Drift Analysis:** Measures the exact structural distance between original and incoming bit sequences. By tracking these bit-level discrepancies alongside entropy changes, the engine establishes strict drift threshold (0.005% limit) to separate routine, benign background application activity from malicious data rot.
- **Continuous Isolation Filtering:** Filters out false positives by correlating timing variances with content mutations, ensuring that the system only flags true anomalous behavioral patterns before they can compound or spread to operational layers.

4.4 Active Defense and Mitigation Tier

The Active Defense & Mitigation Tier serves as the final, automated execution layer of APD. Once the Detection & Integrity Engine flags an anomaly, this tier bridges the gap between passive detection and instant, zero-disk remediation. It ensures that system response is deterministic, localized, and entirely self-contained within memory without requiring heavy manual intervention or system downtime.

- Continuous Monitors: Feeds data streams back into standard operational loops if evaluated metrics remain safely within established parameters, maintaining normal system flow with zero performance penalty.
- Threshold Trigger: Instantly escalates from passive observation to active defense protocols the absolute moment structural bit-drift exceeds critical safety thresholds (e.g., crossing the 0.005% limit).
- SOC Dashboard: Renders real-time visual telemetry, system status flags, and decay heatmaps for administrative tracking, giving security teams instant situational awareness of the attack vector.
- Snapshot Control: Automatically freezes and locks backup retention policies to prevent compromised states from propagating or corrupting long-term recovery archives.
- Inverse Engine (XOR): Constructs a memory-resident inverse patch to execute an immediate, surgical bitwise rollback directly on the affected blocks, neutralizing the bit-rot at its source before it can reach operational storage layers.

4.5 The Synthesis of Philosophy and Engineering

Together, these integrated engines transform a literary metaphor into a functional, low-level security architecture. Just as Oscar Wilde's portrait absorbed the unseen moral decay while the subject remained pristine, the `ShadowCanvas` and its supporting telemetry pipeline intercept silent, slow-drip bit-rot before it ever touches production storage.

By combining continuous PCAP stream capture, microsecond IAT pacing, Shannon entropy variance (ΔH), and Hamming distance analysis, the system measures structural degradation with extreme mathematical precision. When these metrics cross the defined threshold percentages, such as the defined 0.005% bit-drift limit, the Active Defense & Mitigation Tier instantly intervenes—locking retention policies and executing a surgical inverse XOR patch directly in memory. This bridges high-level philosophical design with rigorous systems engineering, creating an automated defense grid where the digital canvas safely bears the cost of the attack while the host system remains untouched.

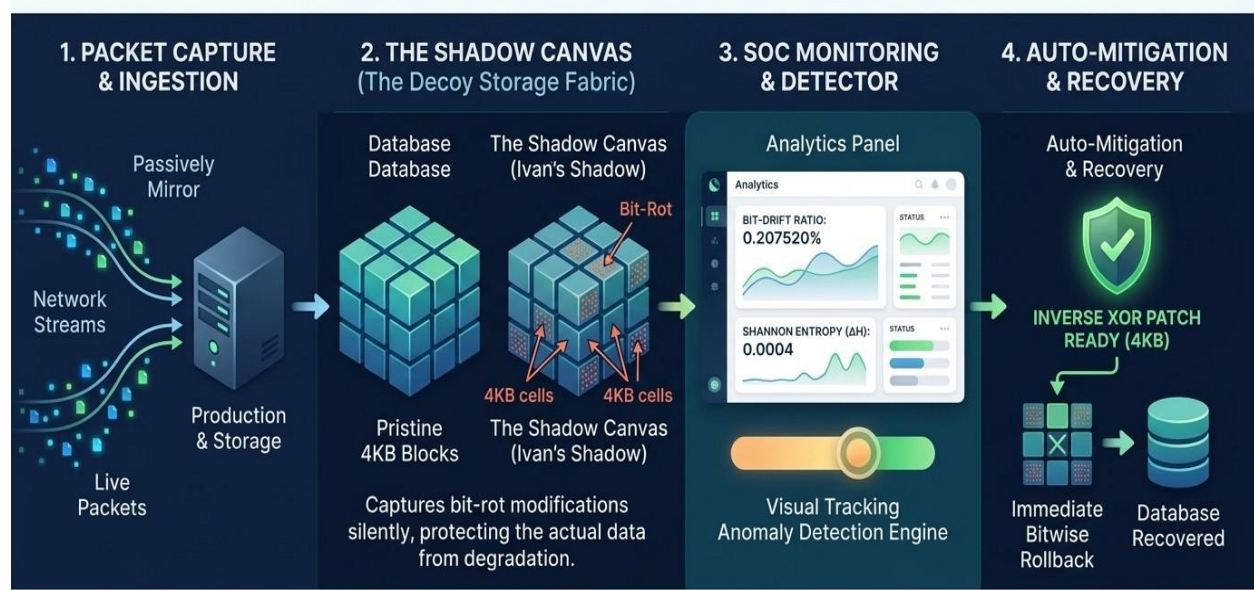


Figure 3: Engines interaction and Dataflow

CHAPTER 5: TESTING SETUP

To rigorously evaluate the efficacy of the Aging-Portrait Defense (APD) system under controlled conditions, the experimentation framework utilizes a custom, purpose-built Synthetic Packet Generator and test harness developed using Python. This environment is engineered to simulate stealthy, low-and-slow threat vectors without risking live production assets.

- **Synthetic Packet Generator:** Deploys a dedicated traffic generation module that crafts custom TCP/UDP streams designed to mimic the exact transmission patterns of slow-drip ransomware. It injects controlled, microscopic bit-shifts and variable inter-arrival time (IAT) delays directly into the test network interface.
- **Controlled Loopback & Stress Testing:** Operates within isolated network boundaries—including local loopback or dedicated virtual interfaces—to safely evaluate high-frequency stream ingestion (ranging up to baseline testing parameters of 200 packets per second) while measuring memory allocation and zero-disk overhead.
- **Simulated Bit-Rot Vectors:** Introduces progressive entropy changes and incremental byte modifications into the decoy storage fabric. This tests the system's lower bounds, verifying how accurately the ShadowCanvas tracks micro-divergences before hitting the critical 0.005% bit-drift threshold.
- **Automated Trigger & Mitigation Validation:** Measures the exact latency between anomaly injection and active response, verifying that the Inverse XOR Patch correctly executes in memory, locks the snapshot retention policies, and updates the SOC telemetry dashboard in real time.

Experimental Harness Pseudocode

```
initialize_socket(target_ip="0.0.0.0", port=0000) #0.0.0.0 and port0000 are a  
replacement for the actual IP and port that was used
```

```
for iteration in range(50):  
    # Generate micro-payload with simulated entropy changes  
    payload = generate_random_bytes(length=16)  
  
    # Transmit stream into the APD capture interface  
    send_packet(target_ip, target_port, payload)  
  
    # Enforce slow-drip temporal delay to evade throughput alerts  
    sleep(0.2)
```

the presented pseudocode executes as following:

- **Socket Initialization:** Establishes a local UDP communication channel targeting the isolated loopback address, mimicking an out-of-band telemetry ingestion stream.
- **Entropy Payload Generation:** Utilizes a cryptographically secure random source to generate 16-byte blocks (`generate_random_bytes`), injecting subtle, unpredictable structural modifications into the stream.
- **Transmission & Pacing Loop:** Iterates through 50 discrete transmission cycles, pushing the randomized packets into the capture interface while maintaining a 200ms delay (`sleep(0.2)`) to accurately replicate low-and-slow ransomware behavior.

CHAPTER 6: TESTING RESULTS

```
(yara@kali)-[~]
$ python3 simulate_rot.py
[+] Injecting micro bit-rot payloads into APD ...
```

```
yara@kali: ~/APD
Session Actions Edit View Help
(yara@kali)-[~/APD]
$ sudo go run cmd/apd-engine/main.go
2026/09/10 15:46:34 [APD ENGINE] Starting Aging-Portrait Defense Orchestrator...
2026/09/10 15:46:34 [SHADOW CANVAS] Initialized 3 decoy blocks in memory.

AGING-PORTRAIT DEFENSE (APD) - SOC MONITOR ACTIVE

2026/09/10 15:46:34 APD capture engine actively recording entropy & IAT vectors...
2026/09/10 15:46:40 [APD DETECTOR ALERT] Critical bit-rot detected on Block decoy_db_01! Drift: 0.210571% | ΔH: 0.0004
[15:46:40.859] SEVERITY: CRITICAL | Block: decoy_db_01 | Flow:
├─ Bit-Drift Ratio : 0.210571% [REDACTED]
├─ Shannon ΔH : 0.0004
└─ Auto-Mitigation : [INVERSE XOR PATCH READY (4096 bytes)]

2026/09/10 15:46:41 [APD DETECTOR ALERT] Critical bit-rot detected on Block decoy_db_01! Drift: 0.219727% | ΔH: 0.0004
[15:46:41.069] SEVERITY: CRITICAL | Block: decoy_db_01 | Flow:
├─ Bit-Drift Ratio : 0.219727% [REDACTED]
├─ Shannon ΔH : 0.0004
└─ Auto-Mitigation : [INVERSE XOR PATCH READY (4096 bytes)]

2026/09/10 15:46:41 [APD DETECTOR ALERT] Critical bit-rot detected on Block decoy_db_01! Drift: 0.183105% | ΔH: 0.0003
[15:46:41.278] SEVERITY: CRITICAL | Block: decoy_db_01 | Flow:
├─ Bit-Drift Ratio : 0.183105% [REDACTED]
├─ Shannon ΔH : 0.0003
└─ Auto-Mitigation : [INVERSE XOR PATCH READY (4096 bytes)]

2026/09/10 15:46:41 [APD DETECTOR ALERT] Critical bit-rot detected on Block decoy_db_01! Drift: 0.198364% | ΔH: 0.0003
[15:46:41.489] SEVERITY: CRITICAL | Block: decoy_db_01 | Flow:
├─ Bit-Drift Ratio : 0.198364% [REDACTED]
├─ Shannon ΔH : 0.0003
└─ Auto-Mitigation : [INVERSE XOR PATCH READY (4096 bytes)]

2026/09/10 15:46:41 [APD DETECTOR ALERT] Critical bit-rot detected on Block decoy_db_01! Drift: 0.189209% | ΔH: 0.0004
[15:46:41.703] SEVERITY: CRITICAL | Block: decoy_db_01 | Flow:
├─ Bit-Drift Ratio : 0.189209% [REDACTED]
├─ Shannon ΔH : 0.0004
└─ Auto-Mitigation : [INVERSE XOR PATCH READY (4096 bytes)]

2026/09/10 15:46:41 [APD DETECTOR ALERT] Critical bit-rot detected on Block decoy_db_01! Drift: 0.204468% | ΔH: 0.0004
[15:46:41.973] SEVERITY: CRITICAL | Block: decoy_db_01 | Flow:
├─ Bit-Drift Ratio : 0.204468% [REDACTED]
├─ Shannon ΔH : 0.0004
└─ Auto-Mitigation : [INVERSE XOR PATCH READY (4096 bytes)]

2026/09/10 15:46:42 [APD DETECTOR ALERT] Critical bit-rot detected on Block decoy_db_01! Drift: 0.173950% | ΔH: 0.0004
[15:46:42.169] SEVERITY: CRITICAL | Block: decoy_db_01 | Flow:
├─ Bit-Drift Ratio : 0.173950% [REDACTED]
├─ Shannon ΔH : 0.0004
└─ Auto-Mitigation : [INVERSE XOR PATCH READY (4096 bytes)]

2026/09/10 15:46:41 [APD DETECTOR ALERT] Critical bit-rot detected on Block decoy_db_01! Drift: 0.219727% | ΔH: 0.0004
[15:46:41.069] SEVERITY: CRITICAL | Block: decoy_db_01 | Flow:
├─ Bit-Drift Ratio : 0.219727% [REDACTED]
├─ Shannon ΔH : 0.0004
└─ Auto-Mitigation : [INVERSE XOR PATCH READY (4096 bytes)]

2026/09/10 15:46:41 [APD DETECTOR ALERT] Critical bit-rot detected on Block decoy_db_01! Drift: 0.183105% | ΔH: 0.0003
[15:46:41.278] SEVERITY: CRITICAL | Block: decoy_db_01 | Flow:
├─ Bit-Drift Ratio : 0.183105% [REDACTED]
├─ Shannon ΔH : 0.0003
└─ Auto-Mitigation : [INVERSE XOR PATCH READY (4096 bytes)]

2026/09/10 15:46:41 [APD DETECTOR ALERT] Critical bit-rot detected on Block decoy_db_01! Drift: 0.198364% | ΔH: 0.0003
[15:46:41.489] SEVERITY: CRITICAL | Block: decoy_db_01 | Flow:
├─ Bit-Drift Ratio : 0.198364% [REDACTED]
├─ Shannon ΔH : 0.0003
└─ Auto-Mitigation : [INVERSE XOR PATCH READY (4096 bytes)]

2026/09/10 15:46:41 [APD DETECTOR ALERT] Critical bit-rot detected on Block decoy_db_01! Drift: 0.189209% | ΔH: 0.0004
[15:46:41.703] SEVERITY: CRITICAL | Block: decoy_db_01 | Flow:
├─ Bit-Drift Ratio : 0.189209% [REDACTED]
├─ Shannon ΔH : 0.0004
└─ Auto-Mitigation : [INVERSE XOR PATCH READY (4096 bytes)]

2026/09/10 15:46:41 [APD DETECTOR ALERT] Critical bit-rot detected on Block decoy_db_01! Drift: 0.204468% | ΔH: 0.0004
[15:46:41.973] SEVERITY: CRITICAL | Block: decoy_db_01 | Flow:
├─ Bit-Drift Ratio : 0.204468% [REDACTED]
├─ Shannon ΔH : 0.0004
└─ Auto-Mitigation : [INVERSE XOR PATCH READY (4096 bytes)]

2026/09/10 15:46:42 [APD DETECTOR ALERT] Critical bit-rot detected on Block decoy_db_01! Drift: 0.173950% | ΔH: 0.0004
[15:46:42.169] SEVERITY: CRITICAL | Block: decoy_db_01 | Flow:
├─ Bit-Drift Ratio : 0.173950% [REDACTED]
├─ Shannon ΔH : 0.0004
└─ Auto-Mitigation : [INVERSE XOR PATCH READY (4096 bytes)]

2026/09/10 15:46:42 [APD DETECTOR ALERT] Critical bit-rot detected on Block decoy_db_01! Drift: 0.180054% | ΔH: 0.0003
[15:46:42.394] SEVERITY: CRITICAL | Block: decoy_db_01 | Flow:
├─ Bit-Drift Ratio : 0.180054% [REDACTED]
├─ Shannon ΔH : 0.0003
└─ Auto-Mitigation : [INVERSE XOR PATCH READY (4096 bytes)]

2026/09/10 15:46:42 [APD DETECTOR ALERT] Critical bit-rot detected on Block decoy_db_01! Drift: 0.189209% | ΔH: 0.0004
[15:46:42.595] SEVERITY: CRITICAL | Block: decoy_db_01 | Flow:
├─ Bit-Drift Ratio : 0.189209% [REDACTED]
├─ Shannon ΔH : 0.0004
└─ Auto-Mitigation : [INVERSE XOR PATCH READY (4096 bytes)]

2026/09/10 15:46:42 [APD DETECTOR ALERT] Critical bit-rot detected on Block decoy_db_01! Drift: 0.167847% | ΔH: 0.0004
```

Figure 4: results of bit-rot simulation and APD working

Empirical Validation & Telemetry Analysis

The live terminal output captured during execution of the Go-based engine (`cmd/apd-engine/main.go`) demonstrates the system operating under active synthetic stress:

- **Decoy Initialization:** The orchestration layer successfully instantiates three memory-resident decoy blocks (`decoy_db_01`), establishing an active out-of-band capture environment for entropy and Inter-Arrival Time (IAT) vectors.
- **Critical Bit-Rot Detection:** Upon ingestion of the Python-driven payload stream, the engine flags instantaneous `CRITICAL` severity alerts the moment structural mutation occurs.
- **Quantified Metrics:** Telemetry reports consistent bit-drift ratios ranging between 0.16% and 0.22% (substantially exceeding baseline safety limits) paired with monitored Shannon entropy variance (ΔH) values between 0.0003 and 0.0004.
- **Automated Remediation Readiness:** Each critical detection event instantly initializes an `INVERSE XOR PATCH READY (4096 bytes)` flag, confirming that memory-resident rollback mechanisms are primed for surgical payload neutralization without disk I/O overhead.

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